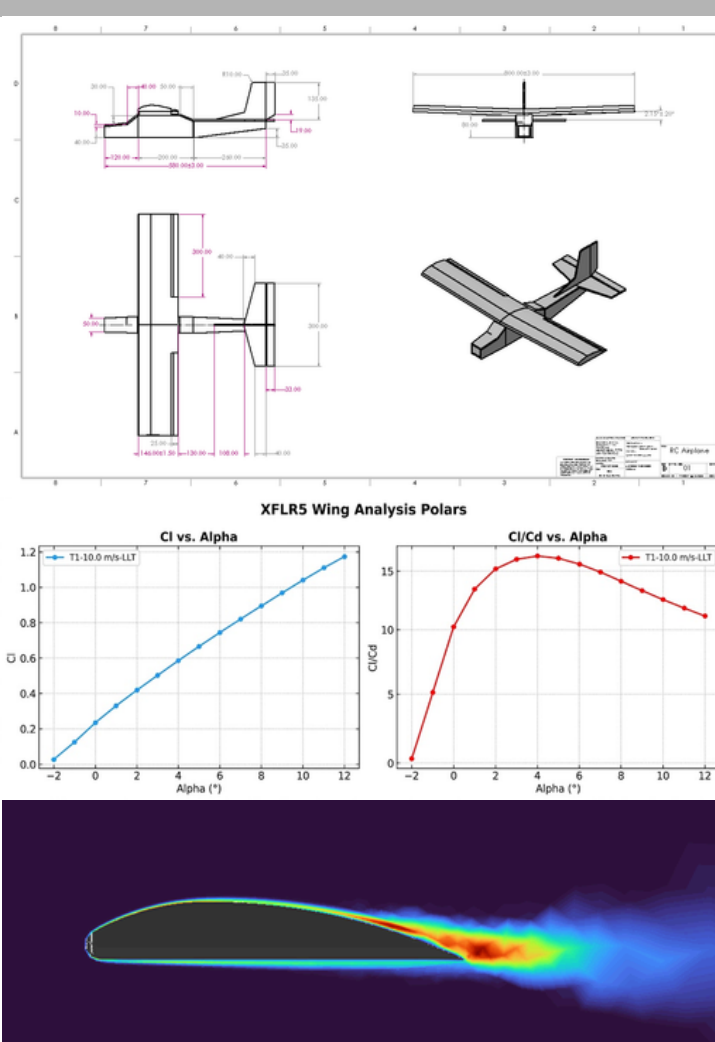


RC Aircraft Design, CFD Analysis & Fabrication



Problem

- Design, build, and fly a 0.6 kg Cessna 172-inspired RC trainer solo, with a 10 m/s cruise at $Re \approx 100k$.
- Pick an airfoil and wing so cruise CL sits near the best-L/D point, using a planform simple enough to hand-build in foam.

Approach

- Designed the full airframe and drawings from scratch in SolidWorks.
- Selected a Clark Y airfoil (airfoiltools.com) for low-Re performance and simple geometry. Ran XFOIL viscous polars at $Re = 50k$, $100k$, and $150k$.
- Ran 3D lifting-line analysis in XFLR5 on the 0.79 m rectangular wing (AR 5.2) at 10 m/s.
- Cross-checked the wing in ANSYS Fluent.
- Hand-built the airframe from EPS foam with a laser-cut motor mount (SolidWorks $\rightarrow$ DXF $\rightarrow$ hardboard).

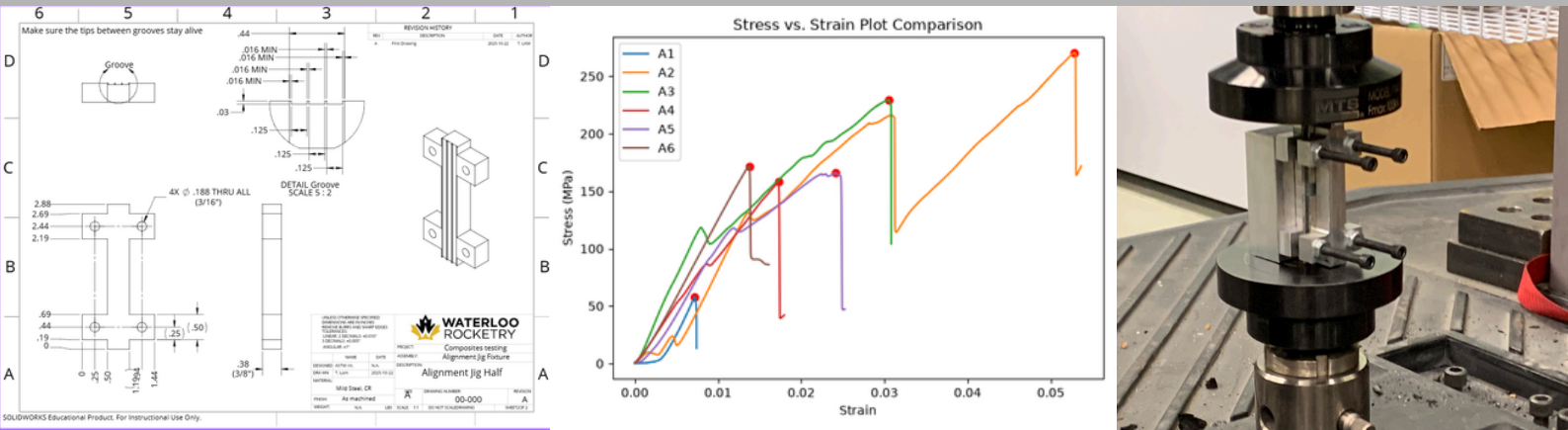
Result

- Wing peak $L/D \approx 15.0$ at $\alpha = 4^\circ$ ($CL \approx 0.57$). $L/D \approx 14$ at the 10 m/s cruise CL of 0.80.
- Found a laminar separation bubble at $Re = 50k$ that cuts peak CL/CD from ≈ 53 ($Re = 100k$) to ≈ 29 .
- Flew 12 flights (~1 hr total). Fixed motor-mount vibration with a second hardboard mount and reduced control over-sensitivity with 30% transmitter expo.



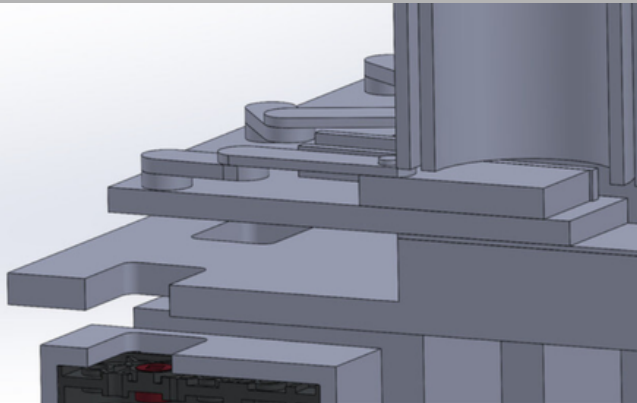
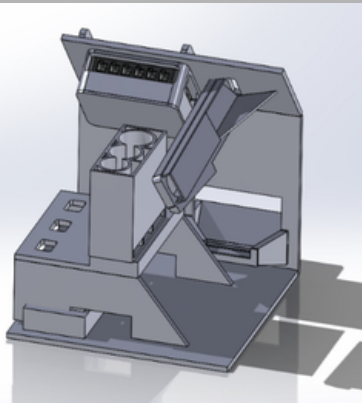
Problem	Approach	Result
- Design a lightweight landing gear for Pegasus 1 that stays structurally sound under landing loads. - Add a controlled breakaway feature so the joint fails before the airframe in a hard crash.	- Designed the full assembly in SolidWorks from 22 mm carbon-fiber tubes and a 3D-printed PETG-CF bracket (127.8 g per leg). - Modeled the bracket alone and ran orthotropic linear-static FEA in ANSYS across a 1–4g load sweep. - Refined the mesh at the bolt interfaces and breakaway neck to confirm convergence.	3.81 factor of safety at 1g landing loads. - Predicted tensile failure at the hourglass neck at 4g, so the joint fails before the tubes or airframe. - Peak stress changed only 1.22% (37.33 to 37.79 MPa) on mesh refinement.

Coupon Compression Testing — Waterloo Rocketry



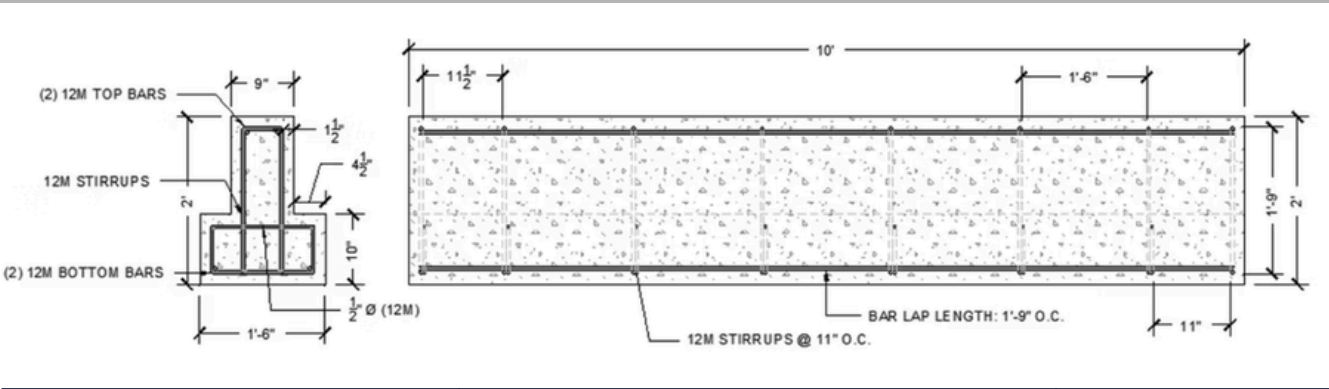
Problem	Approach	Result
- Choose a fiberglass laminate for the Polaris rocket airframe based on compression performance and material cost. - Compare six laminate configurations under standardized testing.	- Machined a Boeing BSS 7260 fixture and made CNC-waterjet aluminum alignment tabs. - Compression-tested six laminates to ASTM D695 / DIN EN 2850 on a 15 kN system. - Wrote a Python pipeline that turns load/displacement data into stress-strain curves and peak stress.	- Peak stress ranged from about 60 to 270 MPa across the six laminates. - The Airstone epoxy laminate was adopted for the Polaris airframe. - Polaris set the amateur liquid-fuel altitude record (63,497 ft, Launch Canada 2026).

VEX IQ Automated Coin Management System



Problem	Approach	Result
• Design a multi-user coin ATM on the VEX IQ platform that detects coins, dispenses withdrawals, and protects user accounts. • Catch jams and block tampering and overdrawing.	• Modeled the enclosure and coin path in SolidWorks, then built it from laser-cut HDF and 3D-printed parts. • Calibrated an optical sensor for coin detection. • Wrote a 700+ line C++ controller: 7 modes at ~50 Hz with PIN login, tamper detection, overdraw prevention and session timeout. • Managed the build and integration for a team of four.	• 96.7% coin-detection accuracy after sensor calibration. • 100% withdrawal accuracy across all test trials. • 0.41 s average jam-detection response. • Verified against ASTM-style test procedures.

Concrete T-Beam Design and Fabrication



Problem	Approach	Result
• Design and build a structurally reinforced concrete beam that uses less concrete than a standard rectangular beam while maintaining load-bearing capacity.	• Selected an inverted T-section to concentrate concrete in high-stress zones. • Engineered tension-zone-only reinforcement with stirrups spaced tighter near the ends per ACI 318-19. • Drafted the section in AutoCAD and built custom cardboard formwork with 1.75 in. laps to hold tolerances during hydration.	• Cut concrete volume by over 20% versus a solid rectangular beam of the same dimensions. • Calculated moment capacity of 1.29 kN·m. • Cast a scale model with the formwork holding its shape and no reinforcement protruding.